Hypothesis Testing

1: What is a null hypothesis (H_0) and why is it important in hypothesis testing?
The null hypothesis (H_0) is a statement that assumes no significant difference, effect, or relationship exists in a population. It is important because it provides a baseline for statistical testing and helps researchers make objective decisions based on sample data.

2: What does the significance level (α) represent in hypothesis testing?
The significance level (α) is the maximum probability of making a Type I error, i.e., rejecting a true null hypothesis. It serves as the threshold for deciding whether the evidence against H_0 is strong enough to reject it. Common values are 0.05, 0.01, and 0.10.

3: Differentiate between Type I and Type II errors.
A Type I error occurs when a true null hypothesis is rejected (false positive), while a Type II error occurs when a false null hypothesis is not rejected (false negative). The probability of a Type I error is denoted by α, and the probability of a Type II error is denoted by β.

4: Explain the difference between a one-tailed and two-tailed test. Give an example of each.
A one-tailed test is used when the alternative hypothesis specifies a direction (e.g., $\mu > 100$ or $\mu < 100$). A two-tailed test is used when the alternative hypothesis only states that the parameter is different ($\mu \neq$ value). For example, testing whether sales increased is a one-tailed test, while testing whether sales changed is a two-tailed test.

5: A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At α = 0.05, test the claim.
Hypothesis Testing (One-Sample t-Test)

The company claims that the average complaint resolution time is 10 minutes.
$H0 : \mu=10$

(This is a two-tailed test because we are testing whether the true mean differs from 10.)

Step 2: Given Data

- Sample size: $n=9$
- Sample mean: $\bar{x}=12$
- Sample standard deviation: $s=3$
- Population mean claimed: $\mu 0=10$
- Significance level: $\alpha=0.05$

Since:

- Population standard deviation is unknown.
- Sample size is small ($n<30$).

We use a t-test.

Step 3: Calculate the Test Statistic
The t-statistic is:
$t=(\bar{x}-\mu_0)/(s/\sqrt{n})$
Substitute the values:
$t=(12-10)/(3/\sqrt{9}) = 2$

Step 4: Determine the Critical Value
Degrees of freedom:
$df=n-1=9-1=8$
For a two-tailed test at $\alpha=0.05$:
$T_{0.025,8} = \pm 2.306$

Step 5: Decision Rule
- Reject H0 if $|t| > 2.306$
- Fail to reject H0 if $|t| \le 2.306$

Our calculated value:
$|t|=2$
Since:
$2<2.306$
we fail to reject H0

At the 5% significance level, there is insufficient evidence to conclude that the average complaint resolution time differs from 10 minutes.
Final Answer
- Test statistic: $t=2.00$
- Critical value: ± 2.306
- Decision: Fail to reject H0

Conclusion: The sample does not provide enough evidence to reject the company's claim that the average complaint resolution time is 10 minutes at $\alpha=0.05$.

6: When should you use a Z-test instead of a t-test?
A Z-test is used when the population standard deviation (σ) is known and/or the sample size is large ($n \ge 30$). A t-test is used when σ is unknown, especially for small samples ($n < 30$). The Z-test uses the normal distribution, while the t-test uses the t-distribution.

7: The productivity of 6 employees was measured before and after a training program. At $\alpha = 0.05$, test if the training improved productivity.

Paired t-Test (Before vs After Training)
Let d=After−Before
H0: $\mu_d = 0$
(No improvement in productivity)

H1: μd>0
(Productivity improved after training)
This is a right-tailed paired t-test.

Step 2: Calculate the Differences

Employee	Before	After	Difference (d)
1	50	55	5
2	60	65	5
3	58	59	1
4	55	58	3
5	62	63	1
6	56	59	3

d=[5,5,1,3,1,3]

d¯=5+5+1+3+1+3/6 = 18/6 = 3

d	d−d¯	(d−d¯)^2
5	2	4
5	2	4
1	-2	4
3	0	0
1	-2	4
3	0	0

∑(d−d¯)^2=16
Sample variance:
sd2=16/6−1 =3.5

t=d¯/sd/sqrt(n)
t≈4.11

Degrees of freedom:
df=n−1=6−1=5
For a one-tailed test at α=0.05:

$t_{0.05,5} = 2.015$

- Calculated $t = 4.11$
- Critical $t = 2.015$

Since
$4.11 > 2.015$
Reject H_0

Conclusion
At the 5% significance level, there is sufficient evidence to conclude that the training program significantly improved employee productivity.
Final Answer

- Mean difference = 3
- Standard deviation of differences = 1.789
- Test statistic = 4.11
- Critical value = 2.015

Reject H0\boxed{\text{Reject } H_0}Reject H0
The training program improved productivity

8: A company wants to test if product preference is independent of gender. At α = 0.05, test independence

We want to test whether product preference is independent of gender.

State the Hypotheses
H_0: Gender and product preference are independent
H_1: Gender and product preference are not independent

Significance level:

$\alpha = 0.05$

Observed Frequencies (O)

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

Expected Frequencies (E)

Formula:

$E = (\text{Row Total})(\text{Column Total}) / \text{Grand Total}$

Male, Product A
$E_{11} = 50 \times 40 / 100 = 20$

Male, Product B
$E_{12} = 50 \times 60 / 100 = 30$

Female, Product A
$E_{21} = 50 \times 40 / 100 = 20$

Female, Product B
$E_{22} = 50 \times 60 / 100 = 30E$

Expected table:

Gender	Product A	Product B
Male	20	30
Female	20	30

7

Formula:

$\chi 2 = \sum (O - E)^2 / E$

Cell	O	E	(O−E)^2/E
Male-A	30	20	5
Male-B	20	30	3.333
Female-A	10	20	5
Female-B	40	30	3.333

$\chi 2 = 5 + 3.333 + 5 + 3.333 = 16.67$

Critical Value

Degrees of freedom:

$df = (r - 1)(c - 1) = (2-1)(2-1) = 1$

At $\alpha = 0.05$ and $df = 1$:

$\chi_{critical}^2=3.841$

Decision

- Calculated $\chi^2 = 16.67$
- Critical $\chi^2=3.841$

Since:

$16.67>3.841$

Reject H_0

Conclusion

There is sufficient evidence at the 5% significance level to conclude that product preference is not independent of gender.

Final Answer
$\chi^2=16.67$
Reject H_0

Conclusion: Gender and product preference are significantly associated. Product preference depends on gender.